

Literature review

Eccentric muscle actions: Implications for injury prevention and rehabilitation

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Abstract

Many acute muscle strain injuries are thought to occur during the eccentric phase of sudden, forceful muscle actions. Repeated eccentric muscle actions during exercise are also thought to contribute to microscopic muscle and tendon damage, leading to chronic muscle strains, muscle rupture and tendinopathy. Conversely, eccentric training has been demonstrated to have a positive effect in the prevention of muscle damage and injury. The properties of eccentric muscle actions which lead to this protective effect remain to be elucidated but are thought to include cellular, mechanical and neural adaptations. This clinical commentary is an attempt to analyze the potential role that eccentric training may have in both the contribution to and prevention of muscle injury by exploring the effect of various parameters on muscle structure and function. Guidelines as to the appropriate design of eccentric training programmes are also provided.

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1. Introduction

Strain injuries occur when muscle and tendon fibres cannot maintain the tension placed upon them and are disrupted. These lesions are among the most common sporting injuries and their rate of recurrence is high (Garrett, 1990; Seward, Orchard, Hazard, & Collinson, 1993), suggesting that current prevention and treatment strategies are less than optimal.

Muscle strain injuries range in severity from minor damage with minimal loss of muscle function, to more complicated conditions with total rupture of muscle continuum, marked reductions in muscle strength, severe bruising and swelling, and severely impaired function (Coburn, 2002).

Many explanations for the production of muscle strain injuries have been offered, however it is likely that a number of factors are involved (Garrett, 1990; Worrell, 1994). Lack of flexibility (McHugh, Connolly, Eston, Kremenec, Nicholas, & Gleim, 1999; Witvrouw, Danneels, Asselman, D'Have, & Cambier, 2003), strength deficits (Croisier, Forthomme, Namurois, Vanderthommen, & Crielaard, 2002; Dauty, Potiron-Josse, & Rochcongar, 2003; Jönghagen, Németh, & Eriksson (1994); Tyler, Nicholas, Campbell, & McHugh, 2001) fatigue (Croisier, 2004; Mair, Seaber, Glisson, & Garrett, 1996), muscle imbalances (Dauty et al., 2003), inadequate warm-up (Evans, Knight, Draper, & Parcell 2002; Hawkins & Fuller, 1999; Safran, Garrett, Seaber, Glisson, & Ribbeck, 1998; Shellock & Prentice, 1985), altered biomechanics (Croisier, 2004) and previous injury (Jönghagen et al., 1994), are among the commonly cited potential causes.

It has been suggested that muscle strains most frequently occur during the eccentric phase of movement, i.e. when the muscle is acting whilst lengthening

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(Garrett, 1990; Kujala, Orava, & Jarvinen, 1997). During eccentric actions, muscles normally act as antagonists counteracting the movement performed by the agonists. For example, towards the end of the recovery phase of the running gait, the hamstring muscles act eccentrically to decelerate forward swing of the lower limb, previously initiated by the agonistic quadriceps (Petersen & Hölmich, 2005). Moreover, exercise that involves eccentric muscle actions is implicated in the causation of exercise-induced muscle damage and soreness (Lieber, Shah, & Fridén, 2002). It has been hypothesized that this type of eccentric-induced muscle damage could be a precursor to macroscopic muscle strain injuries (Brockett, Morgan, & Proske, 2004).

Muscle strains most commonly occur in bi-articular muscles such as the hamstrings, rectus femoris and gastrocnemius (Coburn, 2002). During sporting activities such as sprinting, these long, bi-articular muscles have to cope with high internal forces and rapid changes in muscle length and mode of contraction. An elegant example of this is offered via an videographic report of a cricket player's gastrocnemius being ruptured whilst acting eccentrically to control dorsiflexion of a weight bearing foot (Orchard, Alcott, James, Farhart, Portus, & Waugh, 2002). However, muscle strains have also been reported to occur during slow-lengthening muscle actions such as those performed by ballet dancers (Askling, Tengvar, Saartok, & Thorstensson, 2000).

Conversely, other studies have reported reductions in the incidence of muscle injuries and damage following strengthening and exercise programmes based on eccentric contractions (Askling, Karlsson, & Thorstensson, 2003; Nosaka & Newton, 2002). Similarly, there is support for the use of eccentric exercise programmes in the management of tendinopathy (Peers & Lysens, 2005), especially Achilles tendinopathy (Alfredson, 2003; Ohberg, Lorentzon, & Alfredson, 2004). There is also considerable evidence that eccentric contractions might induce some sort of adaptation which would reduce the likelihood of exercise induced muscle damage. This protective effect has been referred to as the repeated bout effect (Nosaka & Clarkson, 1995). Unfortunately the exact mechanisms regulating this adaptive response are not yet well understood (McHugh, 2003).

This article explores the role of eccentric muscle contractions in the production, prevention and rehabilitation of muscular injuries.

2. Eccentric training and muscle stiffness

Eccentric training produces changes in the normal structure of the muscle–tendon unit. This structure becomes less compliant (stiffer) increasing the force required to produce a change in its length (Leger & Milner, 2000).

Several theories have been postulated in order to explain the increased muscle stiffness observed following eccentric exercise. It has been proposed, for example, that the acute decrease in compliance could be due to an increase in Ca^{2+} released following exercise-induced myofibril disruptions and also as a result of contractures in damaged muscle, i.e. the fibers remaining shortened despite no longer firing impulses are induced (Allen, 2001; Fridén & Lieber, 1992; Jones, Allen, Talbot, Morgan, & Proske, 1997; Whitehead, Weerakkody, Gregory, Morgan, & Proske, 2001). In addition, recent studies have shown that repeated eccentric training leads to longer-term increases in muscle stiffness (Lindstedt, Reich, Keim, & LaStayo, 2002) primarily attributable to adaptation of cytoskeletal proteins such as Actin, Desmin, Titin, Actinin and Nebulin in response to micro-injury (Alter, 2004). These proteins seem to be directly involved in the remodelling process of myofibrils disrupted during eccentric contractions. During this process, the muscle–tendon complex is thought to adapt its viscoelastic properties to enable it to absorb and transmit the forces produced during sudden and forceful contractions (Yu, Fürst, & Thornell, 2003).

However, whether this change in the musculo-tendinous stiffness decreases or increases the risk of muscle injury remains a source of controversy (Shrier, 2002). Recent findings suggest that a stiffer structure could improve the capacity of the muscle-tendon unit to absorb elastic energy, thus improving resistance to disruption (LaStayo, Woolf, Lewek, Reich, & Lindstedt, 2003; Lindstedt, LaStayo, & Reich, 2001; Lindstedt et al., 2002; Reich, Lindstedt, LaStayo, & Pierotti, 2000). These results apparently challenge the role of flexibility programmes aimed at preventing muscle injuries.

From a more classical stand point, others claim that increasing tendon stiffness could augment the risk of injury to the relatively compliant muscular component of the musculo-tendinous unit (Lieber, Leonard, & Brown-Maupin 2000). Supporting this last view McHugh et al. reported a positive correlation between hamstring stiffness and muscle damage following bouts of eccentric muscle actions (McHugh et al., 1999). Others, however, have failed to demonstrate a clear relationship between flexibility and muscle strain injury (Andersen, 2005).

New findings seem to indicate that it is possible that long term eccentric training leads to increased range of motion (ROM) measured in passive conditions (Nelson & Bandy, 2004). Although this idea requires further investigation, it could mean that the increase in passive muscle stiffness observed after chronic eccentric training (Leger & Milner, 2000; Lindstedt et al., 2001, 2002; Reich et al., 2000) does not limit per se, the ROM of a given joint and more importantly, that this adaptation might have prophylactic rather than harmful effects.

This might explain why stretching protocols aimed at reducing muscle damage after eccentric training have not been proved effective (Herbert & Gabriel, 2002; Lund, Vestergaard-Poulsen, Kanstrup, & Sejrsen, 1998).

Muscle length may be directly related to the amount of damage produced after eccentric muscle actions as it seems that the longer the sarcomere length, the greater the damage produced after a single eccentric action (Morgan & Proske, 2004). However, eccentric muscle training performed at short muscle lengths provide little or no “repeated bout” protective effect (McHugh & Pasiakos, 2004; Nosaka, Newton, Sacco, Chapman, & Lavender, 2005) whilst, eccentric muscle actions performed in the outer range of muscle lengths produce a stronger protective effect (Nosaka et al., 2005).

Therefore, it could be suggested that passive muscle stiffness is not the main factor regulating muscle damage susceptibility from eccentric training. Moreover, given that during eccentric actions, muscle contracts while being lengthened, it is likely that functional adaptations might explain these apparently conflicting results.

3. Eccentric training and new sarcomere formation

An increase in the number of sarcomeres distributed in series has been observed following intensive periods of eccentric training (Butterfield, Leonard, & Herzog, 2005; Lynn, Talbot, & Morgan, 1998; Yu et al., 2003). This process of ‘sarcomerogenesis’ is not yet fully understood (Butterfield & Herzog, 2006) but has been suggested to have a protective effect on muscle-tendon structure (Fridén, 1984; Proske & Morgan, 2001) possibly by allowing a reduced sarcomere length at a specific joint angle (Morgan & Proske, 2004). This, in turn, would increase the optimum torque angle, defined as the joint angle at which maximum muscle force is produced (Brockett, Morgan, & Proske, 2000; Proske & Morgan, 2001).

The relationship between sarcomere shortening and the decreased risk of injury is based on the premise that the lower the joint angle at which maximum force is generated, the greater the range through which the muscle is relatively weak, increasing susceptibility to exercise-induced muscle damage and muscle strain injury (Brockett et al., 2004; Gleeson, Eston, Marginson, & McHugh, 2003; Ploutz-Snyder, Tesch, & Dudley, 1998). Illustrating this, Proske and colleagues analyzed the torque-angle curves for human hamstring muscles, before and after bouts of eccentric exercise (Proske, Morgan, Brockett, & Percival, 2004). Apart from observing the shift in optimum torque angle mentioned above, the authors concluded that eccentric actions may protect muscle integrity by increasing the optimum torque angle as a consequence of the addition of new

sarcomeres in series. In a previous study, the same group of researchers reported a reduction of the optimum torque angle in previously injured hamstrings compared with the uninjured muscles (Brockett et al., 2004). Interestingly, in this study, some preliminary data showed that training protocols emphasizing eccentric contractions could significantly reduce the number of hamstring strains injuries.

Interestingly, concentric muscle training appears to reduce the number of muscle sarcomeres in series (Huijing & Jaspers, 2005), which could effectively lower the optimum torque angle. Although controversial (Nosaka, Sakamoto, Newton, & Sacco, 2001), this potential adverse effect of concentric muscle actions should be considered when planning training and rehabilitation programmes.

It seems that not all muscles adapt to training in the same way (Huijing & Jaspers, 2005). Other factors intrinsically related to muscle architecture such as cross-sectional area (CSA), angle of pennation and fibre length could also contribute to varied sarcomere adaptation observed following similar training programmes (Lieber & Fridén, 2000). It is likely that these differences in muscle structure could explain variable susceptibility to injury across muscle groups (Garrett, Nikolaou, Ribbeck, Glisson, & Seaber, 1988) and also equivocal outcomes of eccentric based protocols in the treatment of tendon disorders (Jonsson & Alfredson, 2005; Visnes, Hoksrud, Cook, & Bahr, 2005).

4. Eccentric training and fatigue

There is evidence to suggest that fatigued muscles are less effective in their ability to absorb energy during eccentric contractions (Mair, Seaber, Glisson, & Garrett, 1996). In addition, an epidemiological investigation reported an increased incidence of muscle strain injuries occurring towards the end of either half of football matches (Woods, Hawkins, Maltby, Hulse, Thomas, & Hodson, 2004). Taken together, these findings suggest that fatigued muscles may be predisposed to strain injury.

Surprisingly, eccentric contractions seem to be less influenced by fatigue than other types of muscle actions (Tesch, Dudley, Duvoisin, Hather, & Harris, 1990a). In contrast to concentric muscle actions, fast twitch fibers and high threshold motor units are preferentially recruited during eccentric muscle activity (Nardone, Romanò, & Schieppati, 1989). In addition, levels of motor unit activation are lower during eccentric contractions (Tesch et al., 1990).

How then is it possible that, in spite of preferential recruitment of low endurance, high threshold motor units, eccentric actions display a greater relative resistance to fatigue? It is likely that the mechanical

advantages of lengthening contractions, such as a greater energy absorption and storage may explain this phenomenon (Lastayo et al., 2003).

Several studies have reported that eccentric muscle activity is more efficient than concentric actions at a given metabolic level, as increases in oxygen consumption during eccentric actions are insignificant in comparison with concentric or isometric actions (Dudley, Tesch, Harris, Golden, & Buchanan, 1991; Hather, Tesch, Buchanan, & Dudley, 1991). These results are consistent with investigations illustrating a relatively low ATP turnover (Ryschon, Fowler, Wysong, Anthony, & Balaban, 1997) and a reduced production of ammonia and lactate during eccentric, compared to concentric, muscle actions (Horstmann, Mayer, Maschmann, Niess, Roecker, & Dickhuth, 2001). These findings support the idea of superior metabolic efficiency of eccentric muscle actions.

However, several studies have reported important strength impairments immediately after eccentric exercise (Clarkson, Nosaka, & Braun, 1992). These deficits have been more evident in upper-limb muscles (Byrne, Twist, & Eston, 2004), in females (Sayers & Clarkson, 2001), at short versus long muscle lengths (Child, Saxton, & Donnelly, 1998), and at higher angular velocities of limb movement (Fridén, Sjöström, & Ekblom, 1983). Moreover, it seems that these force disturbances following eccentric actions manifest in any type of muscle action (Michaut, Pousson, Babault, & Van Hoecke, 2002). This force reduction has been attributed to several factors such as excitation–contraction coupling impairment, mechanical adaptations within the muscle (sarcomerogenesis), an increased susceptibility to damage of fast twitch fibers, and impaired glycogen resynthesis (Byrne et al., 2004).

Given that muscle injuries commonly take place when an athlete is fatigued and during eccentric actions it seems logical to train individuals under these specific conditions in order to reduce the risk of muscle damage. However, the side effects of muscle soreness and force disturbances observed after eccentric actions should be considered when planning training programmes.

5. Eccentric training and neural function

Neural adaptations produced by eccentric muscle actions could also contribute to the protective effect that this type of training may induce (McHugh, 2003). These adaptations seem to be related to a more efficient motor unit recruitment patterns produced by eccentric training (Clarkson & Hubal, 2002; Hortobagyi, Houmard, Fraser, Dudek, Lambert, & Tracy, 1998). Enoka suggested that eccentric muscle actions have a specific neural control pathway probably aimed at preventing muscle injuries (Enoka, 1996). Indeed, the unique neural behavior of eccentric actions has been illustrated in

several studies showing: (i) an increased cross-education effect whereby strength gains are produced in the untrained contralateral limb (Enoka, 1996, 1997; Hortobagyi, Lambert, & Hill, 1997); (ii) faster neural adaptation from strength training (Hortobagyi et al. (1996); Hortobagyi, Devita, Money, & Barrier, 2001); (iii) increased cortical activity (Fang, Semionow, Sahgal, Xiong, & Yue, 2001) and (iv) an inverse motor recruitment pattern with preferential recruitment of high threshold motor units at all load levels (Nardone et al., 1989).

It is generally accepted that during eccentric actions there is greater relative force production despite slightly lower surface electromyographic (EMG) activity to comparable concentric muscle actions (Tesch et al., 1990). This fact is likely due to lower relative recruitment and discharge rates of active motor units secondary to the superior mechanical efficiency and energy dissipation of eccentric muscle actions.

Conversely, larger increases in EMG activity have been observed following eccentric, compared to concentric muscle training (Linnamo, Bottas, & Komi, 2000). This muscle activity increase occurs in spite of a decrease in median frequency EMG activity, suggesting that eccentric contractions lead to the enhancement of low-threshold unit activity. In theory, this increased activation of slow muscle fibers after the first bout of eccentric contractions could potentially reduce muscle damage by decreasing the workload on any single muscle fiber (Hortobagyi et al., 1998).

Although there are important differences in the neural control of eccentric versus other types of muscle actions (Enoka, 1996; Bishop, Trimble, Bauer, & Kaminski, 2000), the exact role of the central nervous system in the repeated bout effect is not clear. In fact, some authors have challenged the idea that the protective effect of eccentric training may be explained by neural adaptations. Nosaka and colleagues, for example, showed that the repeated bout effect could also be observed using electrical stimulation, suggesting that the central nervous system is not involved in muscle damage prevention (Nosaka, Newton, & Sacco, 2002a).

6. Eccentric training and proprioception

Proprioception is defined as the capacity of our body to permit the recognition of the position and movements of the limbs in relation with others parts of the body (Gollhofer, 2003). It is thought that proprioception can play an active role in preventing some sporting injuries by enhancing sensory motor control (Verhagen, van der Beek, Twisk, Bouter, Bahr, & van Mechelen, 2004). Proprioceptive information is transmitted by mechanoreceptors mainly located in muscles, tendons, joints and to a lesser extent, in the skin. Whereas position and

movement senses are provided by the muscle spindles, Golgi tendon organs signal changes in the rate of muscle force production (Gollhofer, 2003).

It appears that neuromuscular function can be altered by exercise-induced muscle damage (Bottas, Linnamo, Nicol, & Komi, 2005; Byrne et al., 2004; Deschenes, Brewer, Bush, McCoy, Volek, & Kraemer, 2000). The possibility that changes within muscle structure that result from eccentric training may affect the normal function of these mechanoreceptors has been extensively studied (Brockett, Warren, Gregory, Morgan, & Proske, 1997; Fridén & Lieber, 2001).

Brockett et al. investigated the impact that eccentric and concentric training could have on control of limb position and force perception (Brockett et al., 1997). In this study the subjects sensed that their eccentrically trained arms were more flexed than they actually were whilst concentrically trained arms did not show significant alterations in the perception of limb position. Similar proprioceptive alterations were observed when force sense was assessed. It was concluded that peripheral proprioceptive function could be altered by muscle damage induced by eccentric training.

Previously, Saxton, Clarkson, James, Miles, Westerfer, Clark, Donnelly (1995) showed that the more intense the eccentric exercise performed, the greater the alterations found in the accuracy of proprioception. In this study the altered perception of the eccentrically trained arms in terms of both force and position senses lasted up to three days. This means that alterations in the control of force feedback and limb position can not only be explained as a consequence of acute changes such as inflammation (Whitehead et al., 2001), as the time courses for these two processes do not necessarily coincide.

Gregory, Brockett, Morgan, Whitehead, and Proske (2002) examined the variability in Golgi tendon organs responses to passive and active tension after bouts of eccentric muscle actions (Gregory et al., 2002). Surprisingly, although muscle damage and increases in whole-muscle passive tension were observed, the overall sensitivity of Golgi tendon organs remained unaltered. Furthermore, the same group of researchers (Gregory, Morgan, & Proske, 2003) also reported that muscle spindle function was conserved after eccentric muscle actions thus suggesting that these mechanoreceptors are not prone to be damaged under these specific conditions. Supporting these findings, increases in myotatic reflex, thought to be due to increased muscle spindle activation, have been observed in the patella tendon after bouts of eccentric muscle training (Hortobagyi, Hill, Houmard, Fraser, Lambert, & Israel, 1996).

These findings suggest that impaired sensory-motor sensitivity may not be directly linked to the structural damage observed in muscle structures. In addition, the proprioceptive deficits which appear after performing

eccentric actions can not be explained solely by dysfunction of the peripheral afferent system. It is also evident that the observed increase in tendon organ signalling must have some implications for motor control and therefore, regulatory efferent mechanisms might be involved in these proprioceptive disturbances (Gregory, Morgan, & Proske, 2004; Proske, Weerakkody, Percival, Morgan, Gregory, & Canny, 2003).

Although it is evident that further research is needed to better understand the interactions between eccentric muscle actions and impaired function of the proprioceptive system, in the meantime, the potential for short-term alterations in proprioceptive function should be factored into eccentric training programmes. For example it might be prudent to program (a) low-intensity proprioceptive activities, e.g. single leg balancing and (b) avoid high demand proprioceptive activities e.g. single leg bounding during sessions conducted in the 48 h following eccentric-based training.

7. Eccentric training and muscle strength gains

Many researchers have analyzed the role of eccentric muscle actions in strength training (Dudley, Tesch, Miller, & Buchanan, 1991; Fleck & Kraemer, 2004) however, the effectiveness of this type of training remains controversial (Hortobagyi et al., 2001; Johnson, Adamczyk, Tennoe, & Stromme, 1976). When eccentric training is evaluated using protocols aimed at specifically assessing eccentric or even isometric force, it seems clear that the strength gains obtained are greater than those obtained from performing only concentric actions (Hilliard-Robertson, Schneider, Bishop, & Williams, 2003; Hortobagyi, Hill et al., 1996; Tesch et al., 1990).

Over one and half times maximal concentric load can be moved using eccentric muscle actions (Enoka, 1996). However, current exercise equipment design means that the eccentric load is often limited by the amount of load that can be lifted during the concentric phase of a movement. Therefore, overloading the eccentric phase becomes problematical and is not easily achieved in all muscle groups.

An interesting solution to this problem is offered via the use of a gravity-independent device designed to enhance strength by reducing muscle atrophy and bone mineral loss whilst improving overall fitness during space travel (Alkner & Tesch, 2004; Berg & Tesch, 1994, 1998; Tesch, Ekberg, Lindquist, & Trieschmann, 2004). This equipment utilizes the inertia produced by a rotating flywheel, permitting work in the absence of a gravitational force and allowing the performance of concentric actions while overloading the eccentric phase of movement. Studies conducted with this equipment have demonstrated strength gains, muscle hypertrophy,

and also reductions in muscle strain injuries in elite soccer players (Askling et al., 2003). Moreover, this type of training is thought to permit enhancement of explosive force in a safe way as it allows the performance of plyometric type jumping exercises, also called stretch-shortening-cycle (SSC) exercises, using relatively low jump heights, thereby possibly reducing the impact forces on joints (Komi, 2002).

Nevertheless, given that eccentric training produces muscle damage in direct proportion to the load used during this contraction, the question regarding the optimal intensity for eccentric programmes is not yet clear. Whereas some authors claim that intensity should be high in order to provide the stimulus necessary to produce further adaptations (Hortobagyi, Hill et al., 1996) others have found that the protective effect of eccentric training may be observed even using light resistances (Paddon-Jones & Abernethy, 2001).

8. Relationship between eccentric muscle strength and injury

As force production and magnitude of strain during eccentric muscle actions seem to be directly related to the severity of muscle damage observed after this type of exercise (Armstrong, Warren, & Warren, 1991) many studies have attempted to understand the relationship between strength and injury. Whereas in some studies, the lack of strength has been seen as a possible predictor of strain injuries (Garrett, 1990; Jönghagen et al., 1994; Orchard, Best, & Verrall, 2005; Orchard, Marsden, Lord, & Garlick, 1997), others studies have not found any such correlation (Bennell, Wajswelner, Lew, Schall-Riauour, Leslie, Plant, 1998; Gaida, Cook, Bass, Austen, & Kiss, 2004) even when strength parameters were evaluated after injury (Worrell, 1994).

Brockett et al. (2004), for example, studied whether reductions in isokinetic optimum torque angle of the knee flexors is predictive of hamstring muscle injury. They found that athletes who suffered hamstring muscle strain had lower pre-existing isokinetic force and also a lower optimum torque angle in the injured leg.

Other authors have attempted to use flexor–extensor isokinetic ratios and bilateral flexor ratios as explanations for strain injuries of the knee musculature. Dauty et al. (2003), for example, analyzed whether isokinetic dysfunctions in elite soccer players could be a detector of previous hamstring injuries and a predictor of new muscle strains. In this study only the ratio of eccentric hamstrings to concentric quadriceps strength was useful in predicting previous hamstring injuries, whilst other combinations of parameters did not show any correlation to either previous or future injuries.

Similar studies have reported that athletes who had suffered a muscle strain showed disturbances in force

production (Croisier et al., 2002; Dauty et al., 2003; Jönghagen et al., 1994). These muscle disorders varied depending on the velocity at which isokinetic force was recorded; however, a reduction in eccentric hamstring strength was a common feature in all these studies. Although there are obvious limitations in the use of isokinetic devices to imitate natural human movement, it is likely that using eccentric training regimes to conserve, or even increase optimum torque angle (Proske et al., 2004) and to correct forces imbalances (Croisier et al., 2002), might be an appropriate muscle strain prevention strategy.

9. Guidelines for the design of eccentric training programs

In general, it appears that eccentric training protocols may be beneficial in the prevention and treatment of muscle damage and injury. However, potential adverse effects of this type of training should be considered (Sayers, Clarkson, Rouzier, & Kamen, 1999). The following recommendations may help to maximize the benefits of eccentric training whilst minimizing potentially harmful side effects:

- Given the greater force production during eccentric actions (Enoka, 1996), whenever possible, the use of equipment specifically designed for eccentric training is advocated as these machines allow eccentric overload without high impact forces (Albert, 1995).
- Training parameters should follow the common guidelines applied to any strength or rehabilitation program (Fleck & Kraemer, 2004). However, the volume and intensity of eccentric training programmes should be gradually progressed in order to minimize the effect of exercise-induced muscle damage and to provide the stimulus necessary to produce ongoing adaptations (Cheung, Hume, & Maxwell, 2003; Friedmann et al., 2004).
- If strength gains are desired, eccentric actions should be overloaded from 20% to 80% beyond the maximal isometric strength (Fleck & Kraemer, 2004). However, we recommend the use of highest intensities only with strength athletes familiarized with this type of training.
- When planning the annual training calendar consideration should be given to avoiding high-intensity eccentric exercise during important competition phases as the side effects of transient muscle soreness and strength deficits may impair performance. In addition, the time course for adaptations from eccentric training should also be considered (Albert, 1995; Nosaka et al., 2001).
- As neuromuscular disturbances have been observed during the initial period of eccentric training,

coordination exercises and very demanding technical actions might be best avoided during this phase of training (Byrne et al., 2004).

- Progressively increasing the length at which muscle groups are trained may also help to minimize initial muscle damage whilst gradually improving outer range muscle strength.
- DOMS, along with indirect muscle damage markers such as Creatine Kinase (CK) or Myoglobin levels have a variable response between individuals (Vincent & Vincent, 1997) and therefore they should not be used as a gold standard when training sessions are planned (Nosaka, Newton, & Sacco, 2002b; Salmons, 1997).
- Eccentric actions should be avoided during the initial stages of rehabilitation (Walmsley, Pearson, & Stymiest, 1986). Eccentric protocols should commence as soon as possible (Mannheimer, 1969), normally during the sub-acute phase post-trauma (Albert, 1995). However, the intensity and volume of eccentric based programs should take into consideration aspects such as the degree of injury, the control of inflammation, and the time course adaptations of connective tissue scar formation (Järvinen, Järvinen, Kääriäinen, Kalimo, & Järvinen, 2005).

10. Conclusions

This clinical commentary has analyzed the potential role that eccentric training may have in both the contribution to and prevention of muscle injury by exploring the effect of various parameters on muscle structure and function. Although the exact mechanisms regulating the protective effect observed after eccentric contractions have not yet been elucidated, the use of strength training protocols incorporating lengthening contractions to protect muscle from strain injuries is a promising area of research. Further experimental studies should focus on the exact origin of “the repeated bout effect” in order to design adequate protocols for rehabilitation and prevention of muscle injuries. We recommend the use of randomized control trials (RCT) to analyze the impact of the different types of contractions on muscle injury prevention and rehabilitation. In addition, the use of well-controlled, long-term follow-up studies might bring about some more data related to the incidence of muscle injuries and the factors involved when they occur.

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References

- Albert, M. (1995). *Eccentric muscle training in sports and orthopaedics* (2nd ed). Churchill Livingstone.
- Alfredson, H. (2003). Chronic midportion Achilles tendinopathy: An update on research and treatment. *Clinics in Sports Medicine*, 22(4), 727–741.
- Alkner, B. A., & Tesch, P. A. (2004). Efficacy of gravity-independent resistance device as a countermeasure to muscle atrophy during 29-day bed rest. *Acta Physiologica Scandinavica*, 181, 345–357.
- Allen, D. G. (2001). Eccentric muscle damage: Mechanisms of early reduction of force. *Acta Physiologica Scandinavica*, 171(3), 311–319.
- Alter, M. J. (2004). *Science of flexibility*. 3rd ed. Humankinetics.
- Andersen, J. C. (2005). Stretching before and after exercise: Effect on muscle soreness and injury risk. *Journal of Athletic Training*, 40(3), 218–220.
- Armstrong, R. B., Warren, G. L., & Warren, J. A. (1991). Mechanisms of exercise-induced muscle fiber injury. *Sports Medicine*, 12, 184–207.
- Askling, C., Karlsson, J., & Thorstensson, A. (2003). Hamstring injury occurrence in elite soccer players after preseason strength training with eccentric overload. *Scandinavian Journal of Medicine in Science and Sports*, 13, 244–250.
- Askling, C., Tengvar, M., Saartok, T., & Thorstensson, A. (2000). Sports related hamstring strains—Two cases with different aetiologies and injury sites. *Scandinavian Journal of Medicine in Science and Sports*, 10(5), 304–307.
- Bennell, K., Wajswelner, H., Lew, P., Schall-Riauour, A., Leslie, S., Plant, D., et al. (1998). Isokinetic strength testing does not predict hamstring injury in Australian Rules footballers. *British Journal of Sports Med*, 32(4), 309–314.
- Berg, H. E., & Tesch, P. A. (1994). A gravity-independent ergometer to be used for resistance training in space. *Aviation Space and Environmental Medicine*, 65, 752–756.
- Berg, H. E., & Tesch, P. A. (1998). Force and power characteristics of a resistive exercise device for use in space. *Acta Astronautica*, 42, 1–8.
- Bishop, M. D., Trimble, M. H., Bauer, J. A., & Kaminski, T. W. (2000). Differential control during maximal concentric and eccentric loading revealed by characteristics of the electromyogram. *Journal of Electromyography and Kinesiology*, 10(6), 399–405.
- Bottas, R., Linnamo, V., Nicol, C., & Komi, P. V. (2005). Repeated maximal eccentric actions causes long-lasting disturbances in movement control. *European Journal of Applied Physiology*, 94(1–2), 62–69 Epub 2004 Dec 18.
- Brockett, C. L., Morgan, D. L., & Proske, U. (2000). Human hamstring muscles adapt to eccentric exercise by changing optimum length. *Medicine and Science in Sports and Exercise*, 33(5), 783–790.
- Brockett, C. L., Morgan, D. L., & Proske, U. (2004). Predicting hamstrings strains injury in elite athletes. *Medicine and Science in Sports and Exercise*, 36(3), 379–387.
- Brockett, C. L., Warren, N., Gregory, J. E., Morgan, D. L., & Proske, U. (1997). A comparison of the effects of concentric versus eccentric exercise on force and position sense at the human elbow joint. *Brain Research*, 771, 251–258.
- Butterfield, T. A., & Herzog, W. (2006). The magnitude of muscle strain does not influence serial sarcomere number adaptations following eccentric exercise. *Pflugers Archiv*, 451(5), 688–700 Epub 2005 Aug 23.
- Butterfield, T. A., Leonard, T. R., & Herzog, W. (2005). Differential serial sarcomere number adaptations in knee extensor muscles of rats is contraction type dependent. *Journal of Applied Physiology*, 99, 1352–1358.
- Byrne, C., Twist, C., & Eston, R. (2004). Neuromuscular function after exercise-induced muscle damage. Theoretical and applied implications. *Sports Medicine*, 34(1), 49–69.
- Cheung, K., Hume, P., & Maxwell, L. (2003). Delayed onset muscle soreness: treatment strategies and performance factors. *Sports Medicine*, 33(2), 145–164.

- Child, R. B., Saxton, J. M., & Donnelly, A. E. (1998). Comparison of eccentric knee extensor muscle actions at two muscle lengths on indices of damage and angle-specific force production in humans. *Journal of Sports Sciences*, 16(4), 301–308.
- Clarkson, P. M., & Hubal, M. J. (2002). Exercise-induced muscle damage in humans. *American Journal Physical Medicine and Rehabilitation*, 81(11 Suppl), S52–S69.
- Clarkson, P. M., Nosaka, K., & Braun, B. (1992). Muscle function after exercise-induced muscle damage and rapid adaptation. *Medicine and Science in Sports and Exercise*(5), 512–520.
- Coburn, P. (2002). Posterior thigh pain. In P. Brukner, & K. Khan (Eds.), *Clinical Sports Medicine* (2nd ed, pp. 411–421). McGraw-Hill, pp.
- Croisier, J. L. (2004). Factors associated with recurrent hamstring injuries. *Sports Medicine*, 34(10), 681–695.
- Croisier, J. L., Forthomme, B., Namurois, M. H., Vanderthommen, M., & Crielaard, J. M. (2002). Hamstrings muscle strain recurrence and strength performance disorders. *American Journal of Sports Medicine*, 30(2), 199–203.
- Dauty, M., Potiron-Josse, M., & Rochcongar, P. (2003). Conséquence et prediction des lésions musculaires de ischiojambiers à partir des paramètres isocinétiques concentriques et excentriques du joueur de football professionnel. *Annales de réadaptation de médecine physique*, 46, 601–606 (Article in french).
- Deschenes, M. R., Brewer, R. E., Bush, J. A., McCoy, R. W., Volek, J. S., & Kraemer, W. J. (2000). Neuromuscular disturbance outlasts other symptoms of exercise-induced muscle damage. *Journal of Neurological Sciences*, 174(2), 92–99.
- Dudley, G. A., Tesch, P. A., Harris, R. T., Golden, C. L., & Buchanan, P. (1991). Influence of eccentric actions on the metabolic cost of resistance exercise. *Aviation Space and Environmental Medicine*, 62, 678–682.
- Dudley, G. A., Tesch, P. A., Miller, B. J., & Buchanan, P. (1991). Importance of eccentric actions in performance adaptations to resistance training. *Aviation Space and Environmental Medicine*, 62, 543–550.
- Enoka, R. M. (1996). Eccentric contractions require unique activation strategies by the nervous system. *Journal of Applied Physiology*, 81(6), 2339–2346.
- Enoka, R. M. (1997). Neural adaptations with chronic physical activity. *Journal of Biomechanics*, 30(5), 447–455.
- Evans, R. K., Knight, K. L., Draper, D. O., & Parcell, A. C. (2002). Effects of warm-up before eccentric exercise on indirect markers of muscle damage. *Medicine and Science in Sports and Exercise* 2002, 34(12), 1892–1899.
- Fang, Y., Semionow, V., Sahgal, V., Xiong, F., & Yue, GH. (2001). Greater movement-related cortical potential during human eccentric versus concentric muscle contractions. *Journal of Neurophysiology*, 86, 1764–1772.
- Fleck, S.J., Kraemer, W.J. (2004). Designing resistance training programs. 3rd ed. Humankinetics, 40–50.
- Fridén, J. (1984). Changes in human skeletal muscle induced by long-term exercise. *Cell and Tissue Research*, 236, 365–372.
- Fridén, J., & Lieber, R. L. (1992). Structural and mechanical basis of exercise-induced muscle injury. *Medicine and Science in Sports and Exercise*, 24(5), 521–530.
- Fridén, J., & Lieber, R. L. (2001). Eccentric exercise-induced injuries to contractile and cytoskeletal muscle fibre components. *Acta Physiologica Scandinavica*, 171(3), 321–326.
- Fridén, J., Sjöström, M., & Ekblom, B. (1983). Myofibrillar damage following intense eccentric exercise in man. *International Journal of Sports Medicine*, 4(3), 170–176.
- Friedmann, B., Kinscherf, R., Vorwald, S., Muller, H., Kucera, K., Borisch, S., et al. (2004). Muscular adaptations to computer-guided strength training with eccentric overload. *Acta Physiologica Scandinavica*, 182, 77–88.
- Gaida, J. E., Cook, J. L., Bass, S. L., Austen, S., & Kiss, Z. S. (2004). Are unilateral and bilateral patellar tendinopathy distinguished by differences in anthropometry, body composition, or muscle strength in elite female basketball players? *British Journal of Sports Medicine*, 38(5), 581–585.
- Garrett, W. E., Jr. (1990). Muscle strains injuries: Clinical and basic aspects. *Medicine and Science in Sports and exercise*, 22, 436–443.
- Garrett, W. E., Jr., Nikolaou, P. K., Ribbeck, B. M., Glisson, R. R., & Seaber, A. V. (1988). The effect of muscle architecture on the biomechanical failure properties of skeletal muscle under passive extension. *American Journal of Sports Medicine*, 16(1), 7–12.
- Gleeson, N., Eston, R., Marginson, V., & McHugh, M. (2003). Effects of prior concentric training on eccentric exercise induced muscle damage. *British Journal of Sports Medicine*, 37(2), 119–125 discussion 125.
- Gollhofer, A. (2003). Proprioceptive training: Considerations for strength and power production. In P. Komi (Ed.), *Strength and power in sport* (2nd ed, pp. 331–342). Blackwell Science.
- Gregory, J. E., Brockett, C. L., Morgan, D. L., Whitehead, N. P., & Proske, U. (2002). Effect of eccentric muscle contractions on Golgi tendon organ responses to passive and active tension in the cat. *Journal of Physiology*, 538(1), 209–218.
- Gregory, J. E., Morgan, D. L., & Proske, U. (2003). Tendon organ as monitors of muscle damage from eccentric contractions. *Experimental Brain Research*, 151, 346–355.
- Gregory, J. E., Morgan, D. L., & Proske, U. (2004). Responses of muscle spindles following a series of eccentric contractions. *Experimental Brain Research*, 157, 234–240.
- Hather, B. M., Tesch, P. A., Buchanan, P., & Dudley, G. A. (1991). Influence of eccentric actions on skeletal muscle adaptations to resistance training. *Acta Physiologica Scandinavica*, 143, 177–185.
- Hawkins, R. D., & Fuller, C. W. (1999). A prospective epidemiological study of injuries in four English professional football clubs. *British Journal of Sports Medicine*, 33(3), 196–203.
- Herbert, R. D., & Gabriel, M. (2002). Effects of stretching before and after exercising on muscle soreness and risk of injury: Systematic review. *British Medical Journal*, 325(7362), 468–470.
- Hilliard-Robertson, P. C., Schneider, S. M., Bishop, S. L., & Williams, M. E. (2003). Strength gains following different combined concentric and eccentric exercise regimens. *Aviation Space and Environmental Medicine*, 74(4), 342–347.
- Horstmann, T., Mayer, F., Maschmann, J., Niess, A., Roecker, K., & Dickhuth, H. H. (2001). Metabolic reaction after concentric and eccentric endurance-exercise of the knee and ankle. *Medicine and Science in Sports and Exercise*, 33(5), 791–795.
- Hortobagyi, T., Barrier, J., Beard, D., Braspeninx, J., Koens, P., Devita, P., et al. (1996). Greater initial adaptations to submaximal muscle lengthening than maximal shortening. *Journal Applied Physiology*, 81(4), 1677–1682.
- Hortobagyi, T., Devita, P., Money, J., & Barrier, J. (2001). Effects of standard and eccentric overloaded strength training in young women. *Medicine and Science in Sports and Exercise*, 33(7), 1206–1212.
- Hortobagyi, T., Hill, J. P., Houmard, J. A., Fraser, D. D., Dudek, R., Lambert, J., et al. (1998). Normal forces and myofibrillar disruption after repeated eccentric exercise. *Journal of Applied Physiology*, 84(2), 492–498.
- Hortobagyi, T., Hill, J. P., Houmard, J. A., Fraser, D. D., Lambert, J., & Israel, R. G. (1996). Adaptive responses to muscle lengthening and shortening in humans. *Journal of Applied Physiology*, 80(3), 765–772.
- Hortobagyi, T., Lambert, J., & Hill, J. P. (1997). Greater cross education following training with muscle lengthening than shortening. *Medicine and Science in Sports and Exercise*, 29(1), 107–112.
- Huijing, P. A., & Jaspers, R. T. (2005). Adaptation of muscle size and myofascial force transmission: A review and some experimental

- results. *Scandinavian Journal of Medicine in Science and Sports*, 15, 349–380.
- Järvinen, T. A., Järvinen, T. L., Kääriäinen, M., Kalimo, H., & Järvinen, M. (2005). Muscle injuries: Biology and treatment. *American Journal of Sports Medicine*, 33(5), 745–764.
- Johnson, B. L., Adamczyk, K. J. W., Tennoe, K. O., & Stromme, S. B. (1976). A comparison of concentric and eccentric muscle training. *Medicine and Science in Sports*, 8, 35–38.
- Jonsson, P., & Alfredson, H. (2005). Superior results with eccentric compared to concentric quadriceps training in patients with jumper's knee: A prospective randomised study. *British Journal of Sports Medicine*, 39(11), 847–850.
- Jönhagen, S., Németh, G., & Eriksson, E. (1994). Hamstring injuries in sprinters: The role of concentric and eccentric hamstring muscle strength and flexibility. *American Journal of Sports Medicine*, 22(2), 262–266.
- Jones, C., Allen, T., Talbot, J., Morgan, D. L., & Proske, U. (1997). Changes in the mechanical properties of human and amphibian muscle after eccentric exercise. *European Journal of Applied Physiology*, 76, 21–31.
- Komi, P. V. (2002). *Strength and power in sport* (2nd ed). Blackwell p. 184–202.
- Kujala, U. M., Orava, S., & Jarvinen, M. (1997). Hamstring injuries. Current trends in treatment and prevention. *Sports Medicine*, 23(6), 397–404.
- LaStayo, P. C., Woolf, J. M., Lewek, M. D., Reich, T. E., & Lindstedt, S. L. (2003). Eccentric muscle contractions: Their contribution to injury, prevention, rehabilitation, and sport. *Journal of Orthopaedic and Sports Physical Therapy*, 33, 557–571.
- Leger, A. B., & Milner, T. E. (2000). The effect of eccentric exercise on intrinsic and reflex stiffness in the human hand. *Clinical Biomechanics*, 15, 574–582.
- Lieber, R. L., & Fridén, J. (2000). Functional and clinical significance of skeletal muscle architecture. *Journal of Muscle Nerve*, 23(11), 1647–1666.
- Lieber, R. L., Leonard, M. E., & Brown-Maupin, C. G. (2000). Effects of muscle contractions on the load-strain properties of frog aponeurosis and tendon. *Cell Tissues Organs*, 166(1), 48–54.
- Lieber, R. L., Shah, S., & Fridén, J. (2002). Cytoskeletal disruption after eccentric contraction-induced muscle injury. *Clinical Orthopaedics and related Research*, 403S, S90–S99.
- Lindstedt, S. L., LaStayo, P. C., & Reich, T. E. (2001). When active muscles lengthen: Properties and consequences of eccentric contractions. *News in Physiological Sciences*, 16, 256–261.
- Lindstedt, S. L., Reich, T. E., Keim, P., & LaStayo, P. C. (2002). Do muscles function as adaptable locomotor springs? *Journal of Experimental Biology*, 205, 2211–2216.
- Linnamo, V., Bottas, R., & Komi, P. V. (2000). Force and EMG power spectrum during and after eccentric and concentric fatigue. *Journal of electromyography and Kinesiology*, 10, 293–300.
- Lynn, R., Talbot, J. A., & Morgan, D. L. (1998). Differences in rat skeletal muscles after incline and decline running. *Journal of Applied Physiology*, 85(1), 98–104.
- Lund, H., Vestergaard-Poulsen, P., Kanstrup, I. L., & Sejrnsen, P. (1998). The effect of passive stretching on delayed onset muscle soreness, and other detrimental effects following eccentric exercise. *Scandinavian Journal of Medicine in Science and Sports*, 8(4), 216–221.
- Mair, S. D., Seaber, A. V., Glisson, R. R., & Garrett, W. E., Jr. (1996). The role of fatigue in susceptibility to acute muscle strain injury. *American Journal of Sports Medicine*, 24(2), 137–143.
- Mannheimer, J. S. (1969). A comparison of strength gain between concentric and eccentric contractions. *Physical Therapy*, 49, 1207.
- McHugh, M. P. (2003). Recent advances in the understanding of the repeated bout effect: The protective effect against muscle damage from a single bout of eccentric exercise. *Scandinavian Journal of Medicine in Science and Sports*, 13, 88–97.
- McHugh, M. P., Connolly, D. A. J., Eston, R. G., Kremenec, I. J., Nicholas, S. J., & Gleim, G. W. (1999). The role of passive muscle stiffness in symptoms of exercise-induced muscle damage. *American Journal of Sports Medicine*, 27(5), 594–599.
- McHugh, M. P., & Pasiakos, S. (2004). The role of exercising muscle length in the protective adaptation to a single bout of eccentric exercise. *European Journal of Applied Physiology*, 93, 286–293.
- Michaut, A., Pousson, M., Babault, N., & Van Hoecke, J. (2002). Is eccentric exercise-induced torque decrease contraction type dependent? *Medicine and Science in Sports and Exercise*, 34(6), 1003–1008.
- Morgan, D. L., & Proske, U. (2004). Popping sarcomere hypothesis explains stretch-induced muscle damage. *Clinical and Experimental Pharmacology and Physiology*, 31, 541–545.
- Nardone, A., Romanò, C., & Schieppati, M. (1989). Selective recruitment of high threshold human motor units during voluntary isotonic lengthening of active muscles. *Journal of Physiology*, 409, 451–471.
- Nelson, R. T., & Bandy, W. D. (2004). Eccentric training and static stretching improve hamstring flexibility of high school males. *Journal of Athletic Training*, 39(3), 254–258.
- Nosaka, K., & Clarkson, P. M. (1995). Muscle damage following repeated bouts of high force eccentric exercise. *Medicine and Science in Sports and Exercise*, 27(9), 1263–1269.
- Nosaka, K., & Newton, M. (2002). Concentric or eccentric training effect on eccentric-induced muscle damage. *Medicine and Science in Sports and Exercise*, 34(1), 63–69.
- Nosaka, K., Newton, M., & Sacco, P. (2002a). Responses of human elbow flexor muscles to electrically stimulated forced lengthening exercise. *Acta Physiologica Scandinavica*, 174(2), 137–145.
- Nosaka, K., Newton, M., & Sacco, P. (2002b). Delayed-onset muscle soreness does not reflect the magnitude of eccentric exercise-induced muscle damage. *Scandinavian Journal of Medicine in Science and Sports*, 12(6), 337–346.
- Nosaka, K., Newton, M., Sacco, P., Chapman, D., & Lavender, A. (2005). Partial protection against muscle damage by eccentric actions at short muscle lengths. *Medicine and Science in Sports and Exercise*, 37(5), 746–753.
- Nosaka, K., Sakamoto, K., Newton, M., & Sacco, P. (2001). How long does the protective effect on eccentric exercise-induced muscle damage last. *Medicine and Science in Sports and Exercise*, 33(9), 1490–1495.
- Ohberg, L., Lorentzon, R., & Alfredson, H. (2004). Eccentric training in patients with chronic Achilles tendinosis: Normalised tendon structure and decreased thickness at follow up. *British Journal of Sports Medicine*, 38(1), 8–11 discussion 11.
- Orchard, J. M., Alcot, E., James, T., Farhart, P., Portus, M., & Waugh, S. R. (2002). Exact moment of a muscle strain captured on video. 36, 222–223. www.bjsportsmed.com.
- Orchard, J., Best, T. M., & Verrall, G. M. (2005). Return to play after muscle strains. *Clinical Journal of Sports Medicine*, 15(6), 441–463.
- Orchard, J., Marsden, J., Lord, S., & Garlick, D. (1997). Preseason hamstring muscle weakness associated with hamstring muscle injury in Australian footballers. *American Journal of Sports Medicine*, 25(1), 81–85.
- Paddon-Jones, D. J., & Abernethy, P. J. (2001). Acute adaptation to low volume eccentric exercise. *Medicine and Science in Sports and Exercise*, 33(7), 1213–1219.
- Peers, K. H., & Lysens, R. J. (2005). Patellar tendinopathy in athletes: Current diagnostic and therapeutic recommendations. *Sports Medicine*, 35(1), 71–87.
- Petersen, J., & Hölmich, P. (2005). Evidence based prevention of hamstring injuries in sport. *British Journal of Sports Medicine*, 39(6), 319–323.

- Ploutz-Snyder, L. L., Tesch, P. A., & Dudley, G. A. (1998). Increased vulnerability to eccentric exercise-induced dysfunction and muscle injury after concentric training. *Archives of Physical Medicine and Rehabilitation*, 79(1), 58–61.
- Proske, U., & Morgan, D. L. (2001). Muscle damage from eccentric exercise: Mechanism, mechanical signs, adaptation and clinical application. *Journal of Physiology*, 537(2), 333–345.
- Proske, U., Morgan, D. L., Bocket, C. L., & Percival, P. (2004). Identifying athletes at risk of hamstrings strains and how to protect them. *Clinical and Experimental Pharmacology and Physiology*, 31, 546–550.
- Proske, U., Weerakkody, N. S., Percival, P., Morgan, D. L., Gregory, J. E., & Canny, B. J. (2003). Force-matching errors after eccentric exercise attributed to muscle soreness. *Clinical and Experimental Pharmacology and Physiology*, 30(8), 576–579.
- Reich, T. E., Lindstedt, S. L., LaStayo, P. C., & Pierotti, D. J. (2000). Is the spring quality of muscle plastic? *American Journal Physiology-Regulatory Integrative and Comparative Physiology*, 278, R1661–R1666.
- Ryschon, T. W., Fowler, M. D., Wysong, R., Anthony, A. R., & Balaban, R. S. (1997). Efficiency of human skeletal muscle in vivo: Comparison of isometric, concentric, and eccentric muscle action. *Journal of Applied Physiology*, 83(3), 867–874.
- Safran, M. R., Garrett, W. E., Jr., Seaber, A. V., Glisson, R. R., & Ribbeck, B. M. (1998). The role of warmup in muscular injury prevention. *American Journal of Sports Medicine*, 16(2), 123–129.
- Salmons, S. (1997). *Muscle damage* (2nd ed). Oxford Medical Publications.
- Saxton, J. M., Clarkson, P. M., James, R., Miles, M., Westerfer, M., Clark, S., et al. (1995). Neuromuscular dysfunction following eccentric exercise. *Medicine and Science in Sports and Exercise*, 27(8), 1185–1193.
- Sayers, S. P., & Clarkson, P. M. (2001). Force recovery after eccentric exercise in males and females. *European Journal of Applied Physiology*, 84(1–2), 122–126.
- Sayers, S. P., Clarkson, P. M., Rouzier, P. A., & Kamen, G. (1999). Adverse events associated with eccentric exercise protocols: six case studies. *Medicine and Science in Sports and Exercise*, 31(12), 1697–1702.
- Seward, H., Orchard, J., Hazard, H., & Collinson, D. (1993). Football injuries in Australia at the elite level. *Medical Journal of Australia*, 159(5), 298–301.
- Shellock, F. G., & Prentice, W. E. (1985). Warming-up and stretching for improved physical performance and prevention of sports-related injuries. *Sports Medicine*, 2(4), 267–278.
- Shrier, I. (2002). Does stretching prevent injuries? In D. MacAuley, & T. Best (Eds.), *Evidence-based sports medicine* (pp. 97–116). London: BMJ books.
- Tesch, P. A., Dudley, G. A., Duvoisin, M. R., Hather, B. M., & Harris, R. T. (1990). Force and EMG signal patterns during repeated bouts of concentric or eccentric muscle actions. *Acta Physiologica Scandinavica*, 138, 263–271.
- Tesch, P. A., Ekberg, A., Lindquist, D. M., & Trieschmann, J. T. (2004). Muscle hypertrophy following 5-week resistance training using a non-gravity-dependent exercise system. *Acta Physiologica Scandinavica*, 180, 89–98.
- Tyler, T. F., Nicholas, S. J., Campbell, R. J., & McHugh, M. P. (2001). The association of hip strength and flexibility with the incidence of adductor muscle strains in professional ice hockey players. *American Journal of Sports Medicine*, 29(2), 124–128.
- Verhagen, E., van der Beek, A., Twisk, J., Bouter, L., Bahr, R., & van Mechelen, W. (2004). The effect of a proprioceptive balance board training program for the prevention of ankle sprains: A prospective controlled trial. *American Journal of Sports Medicine*, 32(6), 1383–1384.
- Vincent, H. K., & Vincent, K. R. (1997). The effect of training status on the serum creatine kinase response, soreness and muscle function following resistance exercise. *International Journal of Sports Medicine*, 18(6), 431–437.
- Visnes, H., Hoksrud, A., Cook, J., & Bahr, R. (2005). No effect of eccentric training on jumper's knee in volleyball players during the competitive season: A randomized clinical trial. *Clinical Journal of Sports Medicine*, 15(4), 227–234.
- Walmsley, R. P., Pearson, N., & Stymest, P. (1986). Eccentric wrist extensor contractions and the force-velocity relationship in muscle. *Journal of Orthopaedic & Sports Physical Therapy*, 8, 292.
- Whitehead, N. P., Weerakkody, N. S., Gregory, J. E., Morgan, D. L., & Proske, U. (2001). Changes in passive tension of muscle in humans and animals after eccentric exercise. *Journal of Physiology*, 533(2), 593–604.
- Witvrouw, E., Danneels, L., Asselman, P., D'Have, T., & Cambier (2003). Muscle flexibility as a risk factor for developing muscle injuries in male professional soccer players. A prospective study. *American Journal of Sports Medicine*, 31(1), 41–46.
- Woods, C., Hawkins, R. D., Maltby, S., Hulse, M., Thomas, A., & Hodson, A. (2004). Football Association Medical Research Programme. The Football Association Medical Research Programme: An audit of injuries in professional football—Analysis of hamstring injuries. *British Journal of Sports Medicine*, 38(1), 36–41.
- Worrell, T. W. (1994). Factors associated with hamstring injuries. An approach to treatment and preventative measures. *Sports Medicine*, 17(5), 338–345.
- Yu, J., Fürst, D. O., & Thornell, L. (2003). The mode of myofibril remodelling in human skeletal muscle affected by DOMS induced by eccentric contractions. *Histochemistry and Cell Biology*, 119, 383–393.